

Question ID 66f03086

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: 66f03086

SAT1 M 2.1

71, 72, 73, 76, 77, 79, 83, 87, 93

What is the median of the data shown?

- A. 71
- B. 77
- C. 78
- D. 79

Question ID 87071893

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	

ID: 87071893

SAT1 M 2.2

$x + 40 = 95$

What value of x is the solution to the given equation?

Question ID d2047497

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: d2047497

SAT1 M 2.3

What is the area of a rectangle with a length of **17 centimeters (cm)** and a width of **7 cm**?

- A. **24 cm²**
- B. **48 cm²**
- C. **119 cm²**
- D. **576 cm²**

Question ID 0536ad4f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	

ID: 0536ad4f

SAT1 M 2.4

Which expression is equivalent to $20w - (4w + 3w)$?

- A. $10w$
- B. $13w$
- C. $19w$
- D. $21w$

Question ID 0eae6be1

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 0eae6be1

SAT1 M 2.5

The number y is ~~84~~ less than the number x . Which equation represents the relationship between x and y ?

A. $y = x + 84$

B. $y = \frac{1}{84}x$

C. $y = 84x$

D. $y = x - 84$

Question ID 6d04c89d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	

ID: 6d04c89d

SAT1 M 2.6

The expression $\frac{24}{6x+42}$ is equivalent to $\frac{4}{x+b}$, where b is a constant and $x > 0$. What is the value of b ?

- A. 7
- B. 10
- C. 24
- D. 252

Question ID eaab8bc1

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	

ID: eaab8bc1

SAT1 M 2.7

Out of **300** seeds that were planted, **80%** sprouted. How many of these seeds sprouted?

Question ID 68f2cbaf

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	

ID: 68f2cbaf

SAT1 M 2.8

Ty set a goal to walk at least **24** kilometers every day to prepare for a multiday hike. On a certain day, Ty plans to walk at an average speed of **4** kilometers per hour. What is the minimum number of hours Ty must walk on that day to fulfill the daily goal?

- A. **4**
- B. **6**
- C. **20**
- D. **24**

Question ID 3c4ce699

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	

ID: 3c4ce699

SAT1 M 2.9

If $6 + x = 9$, what is the value of $18 + 3x$?

Question ID 20722644

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 20722644

SAT1 M 2.10

The function f is defined by $f(x) = x^3 + 9$. What is the value of $f(2)$?

- A. 14
- B. 15
- C. 17
- D. 18

Question ID 88e13c8c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 88e13c8c

SAT1 M 2.11

The total cost $f(x)$, in dollars, to lease a car for **36** months from a particular car dealership is given by $f(x) = 36x + 1,000$, where x is the monthly payment, in dollars. What is the total cost to lease a car when the monthly payment is **\$400**?

- A. **\$13,400**
- B. **\$13,000**
- C. **\$15,400**
- D. **\$37,400**

Question ID a73a5c22

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: a73a5c22

SAT1 M 2.12

The function g is defined by $g(x) = 10x + 8$. What is the value of $g(x)$ when $x = 8$?

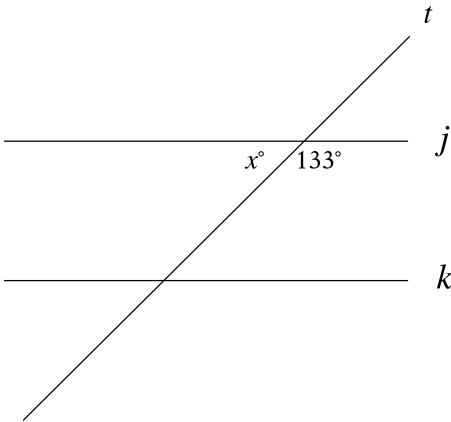
- A. 0
- B. 8
- C. 10
- D. 88

Question ID 3b4b5b1e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 3b4b5b1e

SAT1 M 2.13



Note: Figure not drawn to scale.

In the figure, line *j* is parallel to line *k*. What is the value of *x*?

Question ID 94b48cbf

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	■ ■ ■

ID: 94b48cbf

SAT1 M 2.14

The graph of $7x + 2y = -31$ in the xy -plane has an x -intercept at $(a, 0)$ and a y -intercept at $(0, b)$, where a and b are constants. What is the value of $\frac{b}{a}$?

- A. $-\frac{7}{2}$
- B. $-\frac{2}{7}$
- C. $\frac{2}{7}$
- D. $\frac{7}{2}$

Question ID 7cd1c6db

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 7cd1c6db

SAT1 M 2.15

An object travels at a constant speed of **12** centimeters per second. At this speed, what is the time, in seconds, that it would take for the object to travel **108** centimeters?

- A. **9**
- B. **96**
- C. **120**
- D. **972**

Question ID 6105234d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	

ID: 6105234d

SAT1 M 2.16

John paid a total of **\$165** for a microscope by making a down payment of **\$37** plus p monthly payments of **\$16** each. Which of the following equations represents this situation?

- A. $16p - 37 = 165$
- B. $37p - 16 = 165$
- C. $16p + 37 = 165$
- D. $37p + 16 = 165$

Question ID ba79f10f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	

ID: ba79f10f

SAT1 M 2.17

x	y
0	18
1	13
2	8

The table shows three values of x and their corresponding values of y . There is a linear relationship between x and y . Which of the following equations represents this relationship?

- A. $y = 18x + 13$
- B. $y = 18x + 18$
- C. $y = -5x + 13$
- D. $y = -5x + 18$

Question ID 3918e8bc

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 3918e8bc

SAT1 M 2.18

An object is kicked from a platform. The equation $h = -4.9t^2 + 7t + 9$ represents this situation, where h is the height of the object above the ground, in meters, t seconds after it is kicked. Which number represents the height, in meters, from which the object was kicked?

- A. 0
- B. 4.9
- C. 7
- D. 9

Question ID 044c1cb7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 044c1cb7

SAT1 M 2.19

$h(x) = x^2 - 3$

Which table gives three values of x and their corresponding values of $h(x)$ for the given function h ?

- A.

x	1	2	3
$h(x)$	4	5	6
- B.

x	1	2	3
$h(x)$	−2	1	6
- C.

x	1	2	3
$h(x)$	−1	1	3
- D.

x	1	2	3
$h(x)$	−2	1	3

Question ID 23dedddd

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 23dedddd

SAT1 M 2.20

In the linear function f , $f(0) = 8$ and $f(1) = 12$. Which equation defines f ?

- A. $f(x) = 12x + 8$
- B. $f(x) = 4x$
- C. $f(x) = 4x + 12$
- D. $f(x) = 4x + 8$

Question ID 11ccf3e1

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 11ccf3e1

SAT1 M 2.21

$$14j + 5k = m$$

The given equation relates the numbers j , k , and m . Which equation correctly expresses k in terms of j and m ?

- A. $k = \frac{m-14j}{5}$
- B. $k = \frac{1}{5}m - 14j$
- C. $k = \frac{14j-m}{5}$
- D. $k = 5m - 14j$

Question ID 0e709a29

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	■ ■ ■

ID: 0e709a29

SAT1 M 2.22

$RS = 440$
 $ST = 384$
 $TR = 584$

The side lengths of right triangle RST are given. Triangle RST is similar to triangle UVW , where S corresponds to V and T corresponds to W . What is the value of $\tan W$?

- A. $\frac{48}{73}$
- B. $\frac{55}{73}$
- C. $\frac{48}{55}$
- D. $\frac{55}{48}$